# UNRAMIFIED AND TAMELY RAMIFIED SEPARABLE EXTENSIONS STAY SO AFTER TENSORING WITH AN EXTENSION OF DISCRETE VALUATION RINGS

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*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot; nothing was added, removed or redefined and no package was added.

*Editorial note.* The source URL is the source-level Stacks Project URL, as required when more than one card is produced from one source; the per-claim tag is carried by claim_key.

*Editorial note.* The author writes '(details omitted; hint: use Lemma ...)' at proof line 34363 and '(details omitted)' at proof line 34394; both are kept verbatim and both mark steps left to the reader, the first being the replacement of K' by a larger tame extension and the second the identification of the right-hand ring as a product of Dedekind domains tamely ramified over B.

*Allowed standard prerequisites (editorial).*

- Unramified, tame and the characterisation of tame extensions by adjoining roots of a uniformizer. the two assertions in the statement at lines 34340-34343 and the characterisation invoked at proof lines 34365-34367. Fixes the properties being transported and supplies the reduction to the case where K' contains K[pi^{1/e}].
- Finite etale extensions of a discrete valuation ring and their relation to Dedekind domains, and the integral closure as a finite etale base change. used at proof lines 34353-34359. Is the entirety of case (1) and is reapplied at proof lines 34375-34378.
- Composition of tame extensions. invoked at proof lines 34380-34381. Lets the tameness of the factors L_{e,j} be combined with the unramifiedness obtained from case (1).

*Source reference.* Original lemma 10881: sources/stacks-more-algebra/pinned-original-source.tex, source lines 10881–10881. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

*Source reference.* Original lemma 34185: sources/stacks-more-algebra/pinned-original-source.tex, source lines 34185–34185. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

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*Date*: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

*Source reference.* Original lemma 34138: sources/stacks-more-algebra/pinned-original-source.tex, source lines 34138–34138. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

*Source reference.* Original lemma 34160: sources/stacks-more-algebra/pinned-original-source.tex, source lines 34160–34160. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

**Lemma 0.2.** *Let $A \subset B$ be an extension of discrete valuation rings. Denote $L/K$ the corresponding extension of fraction fields. Let $K'/K$ be a finite separable extension. Then*

$$K' \otimes_K L = \prod L'_i$$

*is a finite product of fields and the following is true*

1. *If $K'$ is unramified with respect to $A$, then each $L'_i$ is unramified with respect to $B$.*
2. *If $K'$ is tamely ramified with respect to $A$, then each $L'_i$ is tamely ramified with respect to $B$.*

*Proof.* The algebra $K' \otimes_K L$ is a finite product of fields as it is a finite étale algebra over $L$. Let $A'$ be the integral closure of $A$ in $K'$.

In case (1) the ring map $A \to A'$ is finite étale. Hence $B' = B \otimes_A A'$ is finite étale over $B$ and is a finite product of Dedekind domains (Lemma 10881). Hence $B'$ is the integral closure of $B$ in $K' \otimes_K L$. It follows immediately that each $L'_i$ is unramified with respect to $B$.

Choose a uniformizer $\pi \in A$. To prove (2) we may replace $K'$ by a larger extension tame ramified with respect to $A$ (details omitted; hint: use Lemma 34160). Thus by Lemma 34185 we may assume there exists some $e \geq 1$ invertible in $\kappa_A$ such that $K'$ contains $K[\pi^{1/e}]$ and such that $K'$ is unramified with respect to $A[\pi^{1/e}]$. Choose a product decomposition

$$K[\pi^{1/e}] \otimes_K L = \prod L_{e,j}$$

For every $i$ there exists a $j_i$ such that $L'_i/L_{e,j_i}$ is a finite separable extension. Let $B_{e,j}$ be the integral closure of $B$ in $L_{e,j}$. By (1) applied to $K'/K[\pi^{1/e}]$ and $A[\pi^{1/e}] \subset (B_{e,j_i})_{\mathfrak{m}}$ we see that $L'_i$ is unramified with respect to $(B_{e,j_i})_{\mathfrak{m}}$ for every maximal ideal $\mathfrak{m} \subset B_{e,j_i}$. Hence the proof will be complete if we can show that $L_{e,j}$ is tamely ramified with respect to $B$, see Lemma 34138.

Choose a uniformizer $\theta$ in $B$. Write $\pi = u\theta^t$ where $u$ is a unit of $B$ and $t \geq 1$. Then we have

$$A[\pi^{1/e}] \otimes_A B = B[x]/(x^e - u\theta^t) \subset B[y,z]/(y^{e'} - \theta, z^e - u)$$

where $e' = e/\gcd(e,t)$. The map sends $x$ to $zy^{t/\gcd(e,t)}$. Since the right hand side is a product of Dedekind domains each tamely ramified over $B$ the proof is complete (details omitted). $\square$